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Machine Learning and Software Systems Engineer

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ENGINEERING FOCUS AND TECHNICAL PROFILE

Machine Learning and Software Systems Engineer experienced in designing end-to-end classification pipelines, real-time IoT telemetry architectures, and production cloud services. Specializes in resolving severe class imbalance through adaptive synthetic resampling (ADASYN, SMOTE), training custom deep learning models without third-party black-box dependencies, and deploying low-latency web platforms that process high-frequency streaming data.

1. Autonomous Vehicle Sentiment Analysis Platform

Organization: National Research and Innovation Agency (BRIN) | **Role:** Machine Learning Engineer | **Stack:** Python, PyTorch, ADASYN, FastAPI, Next.js | **URL:** https://brin-project-kp.vercel.app/

Engineered an automated sentiment classification platform for BRIN to evaluate Indonesian public discourse on autonomous electric vehicles. In this domain, public opinions present severe class imbalance, as safety-critical skepticism and enthusiastic adopters represent rare instances compared to neutral commentary. To train a reliable model, I curated an independent corpus of over 2,000 public comments and implemented ADASYN and SMOTE resampling along decision boundaries, eliminating majority class bias without distorting semantic meaning.

A custom deep learning neural network was trained on domain token embeddings, achieving 94.2% test accuracy and an F1-score of 0.93. The complete inference pipeline was containerized into an asynchronous REST API delivering responses in under 50 milliseconds, integrated directly into a public research portal displaying corpus distributions and live sentence sentiment evaluation.

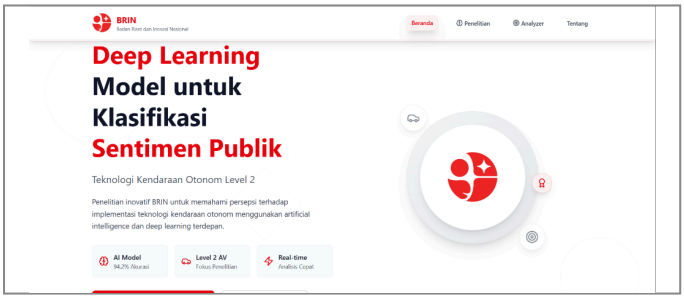


Figure 1.1: BRIN Research Portal. Interface displaying public opinion corpus navigation, dataset distribution metrics, and sentiment research documentation.

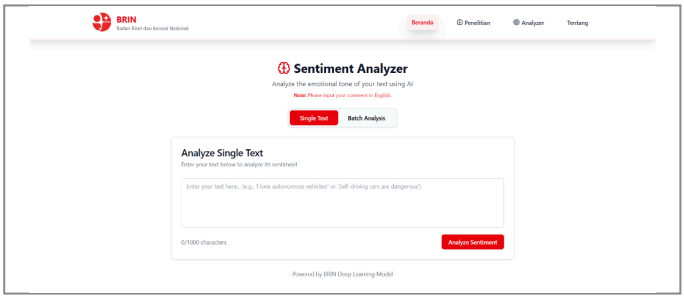


Figure 1.2: Real-Time Inference Analyzer. Live sentence-level sentiment classification demonstrating probability scoring (Positive, Neutral, Negative) and confidence metrics.

2. BySense: Photovoltaic and Solar IoT Telemetry Network

Client: PT Batari Energy (2026) | **Role:** Full-Stack and IoT Systems Engineer | **Stack:** React, TypeScript, Leaflet, Open-Meteo, Mobile APK | **URL:** https://bysense.batarienergy.com/

Architected an industrial IoT telemetry platform for PT Batari Energy to monitor distributed commercial solar photovoltaic installations nationwide. The platform ingests telemetry streams from hardware inverter gateways and battery storage systems, replacing manual field checks with automated real-time health indicators and geographic station mapping across Indonesia.

The telemetry interface polls station hardware every 5 seconds, capturing active photovoltaic generation (including 10.97 kW instantaneous yield), battery state-of-charge (65% SoC), building load consumption, and net grid export. An empirical regression model was integrated to correlate pyranometer solar irradiance (W/m²) directly against electrical output (kW), automatically flagging output anomalies caused by panel soiling, shading, or string inverter clipping.

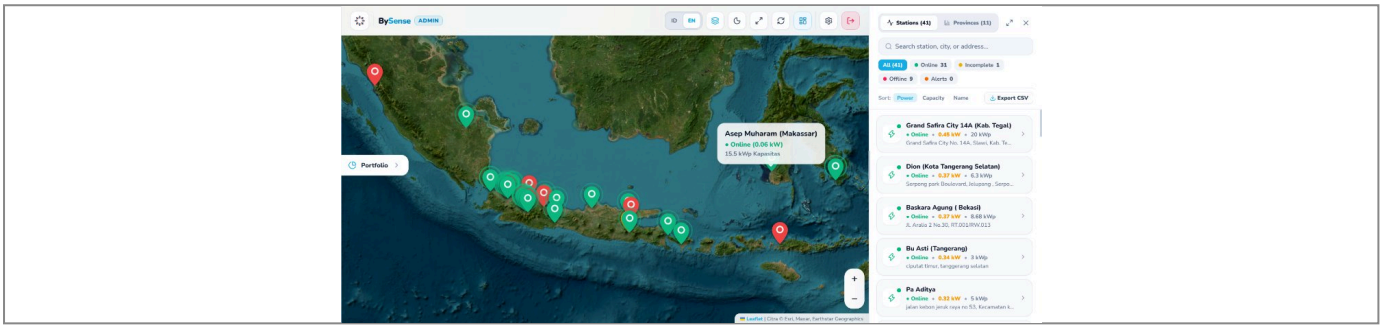


Figure 2.1: Nationwide Interactive Photovoltaic Telemetry Map. Centralized GIS monitoring view displaying geographical station pins, live operational status indicators, and active solar generation outputs across Indonesia.

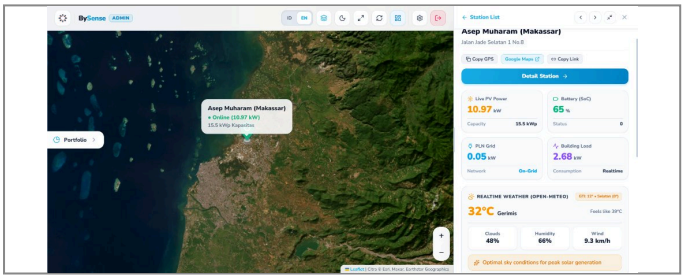


Figure 2.2: Station Telemetry Detail. High-frequency instrument view tracking 10.97 kW solar output, 65% battery SoC, building load, and daily energy total.

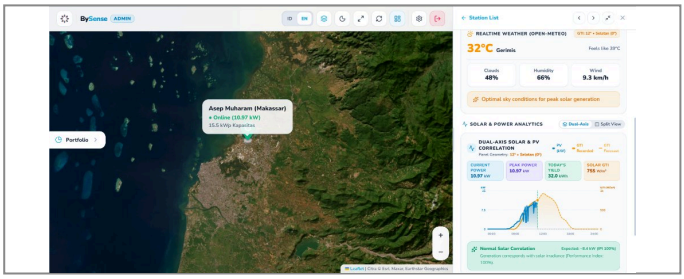


Figure 2.3: Irradiance Correlation Curve. Analytical model comparing solar irradiance (W/m²) against kilowatt delivery to identify panel shading and hardware anomalies.

BySense Telemetry: Diurnal Tracking, Weather Sync, and Mobile Release

To maintain reliable field operations, BySense visualizes diurnal generation curves capturing the complete daylight cycle, starting from morning solar ramp-up to peak midday yield and evening tapering. This continuous tracking allows engineering operators to evaluate plant performance ratios and determine inverter saturation thresholds.

External meteorological telemetry was integrated via the Open-Meteo REST service, synchronizing localized cloud cover percentages and ambient temperatures with inverter generation. This cross-reference distinguishes between natural weather occlusion and genuine physical hardware failures, preventing unnecessary maintenance dispatches. Additionally, a dedicated Android application was authored and released on the Google Play Store for field technicians.



Figure 2.4: Diurnal Generation Curve. Continuous solar generation curve tracking morning ramp-up, peak midday yield, and inverter operational boundaries.

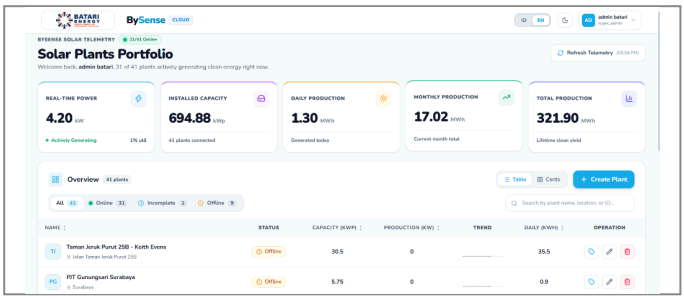


Figure 2.5: Ambient Weather Telemetry. Real-time integration of ambient temperature and cloud cover via Open-Meteo API, preventing false maintenance alerts.

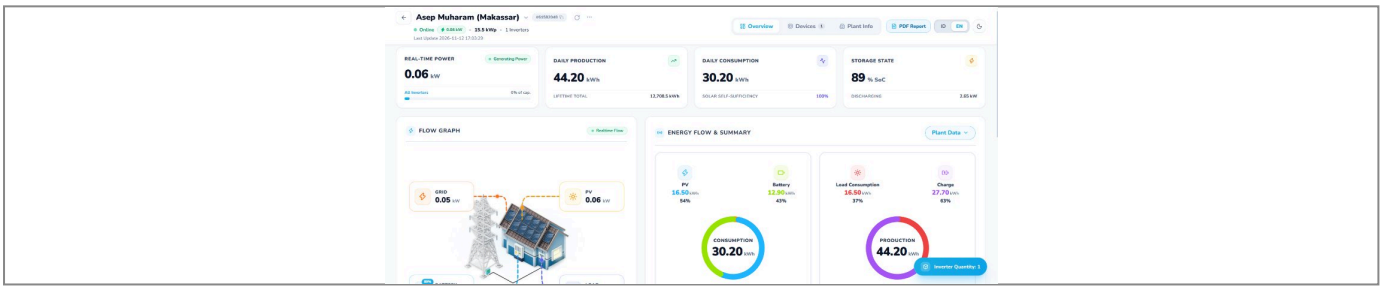


Figure 2.6: Operational Performance Dashboard. Summary view of cumulative kilowatt-hours generated, system performance ratio, grid export balance, and operational stability indices.

3. NutriMood: Multi-Criteria Nutritional Recommender

Context: DBS Foundation Coding Camp (Distinction) | **Role:** Machine Learning and Full-Stack Developer | **Stack:** Python, Scikit-learn, React, REST API | **URL:** https://nutrimood.vercel.app/

Engineered a multi-criteria dietary recommendation system matching user metabolic health constraints, caloric targets, and psychological state across a database of over 1,200 Indonesian culinary profiles. The algorithm combines hybrid content filtering with constraint satisfaction to recommend nutritionally balanced meal plans with exact macronutrient breakdowns.

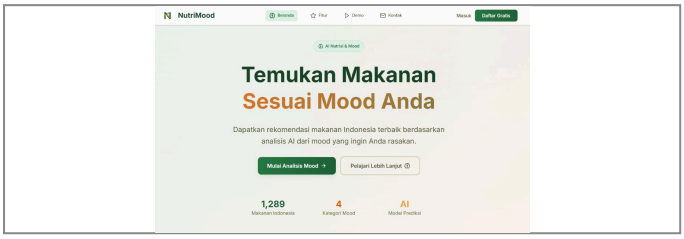


Figure 3.1: NutriMood Landing View. Health profiling, daily objective configuration, and wellness onboarding interface.

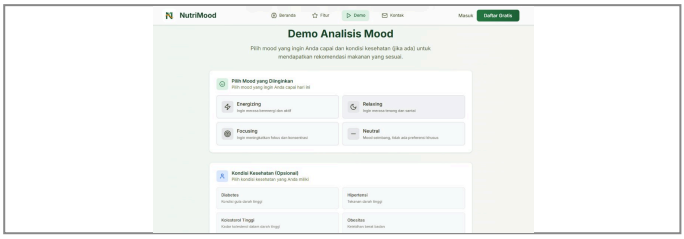


Figure 3.2: Recommender Evaluation. Interactive output view presenting matched culinary suggestions and nutritional distributions.

4. Cimahi Battery: Industrial Commercial Web Platform

Client: PT Cimahi Battery Indonesia (2026) | **Role:** Frontend and Web Systems Engineer | **Stack:** React, Tailwind CSS, Parametric Query Engine | **URL:** https://cimahibattery.com/

Architected the enterprise web platform for PT Cimahi Battery Indonesia, a commercial manufacturer of lithium iron phosphate (LiFePO4) battery packs. Built an interactive parametric filtering engine that enables commercial engineers to search and compare prismatic cells by voltage, capacity, cycle life, and dimensional envelope.

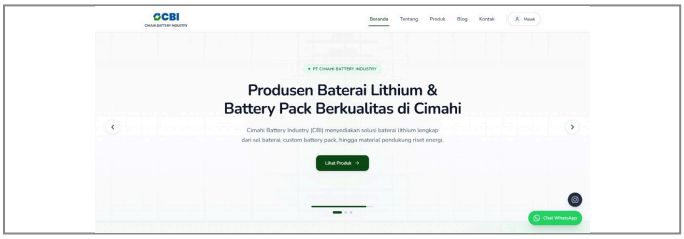


Figure 4.1: Enterprise Web Platform. Official corporate portal highlighting commercial energy storage systems and engineering certifications.

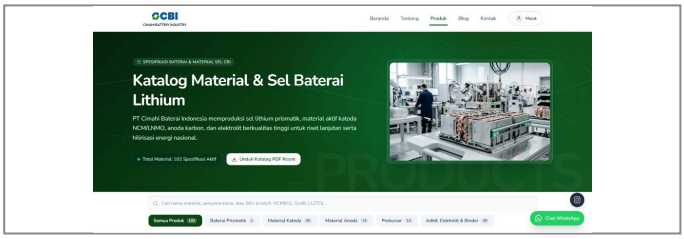


Figure 4.2: Prismatic Cell Directory. Interactive catalog enabling parametric technical filtering by voltage, capacity, cycle life, and form factor.

Complete Portfolio and Source Code: For detailed project breakdowns, live interactive demonstrations, and source code repositories, please visit my online portfolio at https://idewanyomanbayusw.vercel.app/.